

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

S PRAVEEN (192524239)

Signature of the student:

ARTIFICIAL INTELLIGENCE — ASSESSMENT TOOL 3

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1. Question 13 — Online Search Agent

An online search agent operates while it is physically exploring the environment. Unlike an offline search agent, it does not initially know the complete state space. It maintains an internal knowledge structure containing the locations and connections discovered so far, updates that structure after each observation, and chooses its next action using the information currently available.

One possible route and knowledge updates

Step	Agent action / observation	Knowledge after the step
1	Starts at A.	Known: $A \rightarrow B$, $A \rightarrow C$. Goal G is not yet connected.
2	Chooses $A \rightarrow B$ and moves to B.	Known: $A \rightarrow B$, $A \rightarrow C$. At B, it discovers $B \rightarrow D$.
3	Moves $B \rightarrow D$.	Known: $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$. At D, it discovers $D \rightarrow G$.
4	Moves $D \rightarrow G$.	Known: $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$, $D \rightarrow G$. Goal G reached.

Trace: $A \rightarrow B \rightarrow D \rightarrow G$. The important feature is that the agent does not assume that $B \rightarrow D$ or $D \rightarrow G$ was known at the beginning. Those links are added to its internal map only after the corresponding locations are visited and the connections are observed. The alternative edge $A \rightarrow C$ remains known but is not selected in this particular run because the discovered route through B reaches the goal directly.

Conclusion: This demonstrates the core idea of online search: action and learning occur together. The agent repeatedly observes the current state, updates its model of the environment, and then selects the next action based on the improved knowledge.

2. Question 14 — Constraint Satisfaction Problem

This scheduling problem is a Constraint Satisfaction Problem (CSP), because a solution must assign a value from a finite domain to every variable while satisfying all stated constraints.

Variables and domains

Student / Variable	Domain	Constraints
A	9 AM / 10 AM / 11 AM	$A \neq 9 \text{ AM}$
B	9 AM / 10 AM / 11 AM	B must be before C
C	9 AM / 10 AM / 11 AM	$C \neq 11 \text{ AM}$
D	9 AM / 10 AM / 11 AM	$D \neq A$

Constraints

1. A cannot be assigned to 9 AM.
2. B must be scheduled before C.
3. D cannot be scheduled at the same time as A.
4. C cannot be assigned to 11 AM.

One valid assignment

Student	Domain	Assigned Slot
A	9 / 10 / 11 AM	10 AM

Student	Domain	Assigned Slot
B	9 / 10 / 11 AM	9 AM
C	9 / 10 / 11 AM	10 AM
D	9 / 10 / 11 AM	11 AM

Verification: A = 10 AM satisfies $A \neq 9$ AM. B = 9 AM is before C = 10 AM. D = 11 AM is different from A = 10 AM. C = 10 AM satisfies $C \neq 11$ AM. Therefore, all constraints are satisfied and the assignment is valid.

3. Question 15 — Map Colouring CSP

The four regions form the cycle A–B–D–C–A. Adjacent regions must have different colours. The available colours are Red (R), Green (G), and Blue (B). A backtracking method assigns a colour, checks consistency with already assigned neighbours, and backtracks whenever a conflict occurs.

Backtracking steps

Step	Region	Available colours	Decision / result
1	A	R / G / B	Assign A = R.
2	B	G / B	B is adjacent to A, so R is unavailable. Assign B = G.
3	C	G / B	C is adjacent to A, so R is unavailable. Assign C = G.
4	D	R / B	D is adjacent to B and C, both G. Assign D = R.
5	Check	—	A–B, A–C, B–D and C–D all have different colours.

Region	Available Colours	Assigned Colour
A	R / G / B	Red (R)
B	R / G / B	Green (G)
C	R / G / B	Green (G)
D	R / G / B	Red (R)

Why backtracking succeeds: The first choice A = R restricts both B and C from using R. Choosing B = G and C = G is valid because B and C are not adjacent to each other. D is adjacent to both B and C, so it cannot use G; Red is available because D is not adjacent to A. Thus D = R completes the colouring without conflict. The third colour Blue is not required because the cycle is even and can be coloured with two colours.

Overall conclusion: Online search is useful when the environment is initially unknown and information must be learned during movement. CSP provides a systematic way to represent variables, domains, and constraints, while backtracking efficiently explores assignments and rejects inconsistent choices early. These techniques allow AI systems to solve problems where the complete solution is not obtained by simple direct calculation.